

# Supplementary Material for: When Does Population-Mobility OD Injection Help Traffic Forecasting? A Capacity-Gap Account and a Validation-Gated Remedy

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This document contains Appendices A–E referenced from the main paper. Table and figure numbering continues from the main paper (Table 15 / Figure 8 are the main paper’s last table and figure), so table/figure references below match those used in the main text.

## APPENDIX A

### ABLATION STUDY ON THE FUSION ARCHITECTURE

Four architectural embellishments to the fusion mechanism were tested on the lightweight model and, following early evidence of a consistent negative pattern, stopped after 5 folds per a pre-registered “stop early if it is getting worse” policy agreed upon before the sweep began, to avoid spending compute chasing a direction the early folds already argued against.

TABLE 16

Ablation: every tested variant ends up worse than the simple per-node/per-step gate of Section 4.3, most substantially so.

| Variant            | Fold | MSE vs. baseline |
|--------------------|------|------------------|
| +gate-calendar     | 0    | +7.00%           |
| +encoder injection | 0    | +5.03%           |
| +both above        | 0    | +4.31%           |
| +gate-calendar     | 5    | −2.38%           |
| 2-channel (in+out) | 0    | +21.30%          |
| 2-channel (in+out) | 5    | +7.49%           |
| 2-channel (in+out) | 10   | +4.60%           |

Six of the seven tested points are worse than the simple baseline fusion of the main paper’s Section 4.3; the 2-channel variant, which adds a second, inflow-weighted routing channel alongside the outflow channel, is consistently the worst (up to +21% worse), suggesting that adding raw

feature dimensionality to the injected signal without a correspondingly principled reason to do so mainly adds noise and optimization difficulty rather than useful information. The single exception (calendar-in-gate at fold 5, −2.38%) was not judged sufficient to overturn the pattern across the other six points, and the original, simpler design was retained for all subsequent experiments in this paper.

## APPENDIX B

### CAN A MORE PRINCIPLED THRESHOLD DO BETTER THAN THE NAIVE RULE?

GWNET/volume’s reversal suggests an obvious fix: require a sensor’s validation margin to exceed some noise-informed threshold before trusting it, rather than selecting on any margin however small. A 2-combination pilot (GMAN/volume, GWNET/volume) using a fixed quantile of each architecture’s seed-noise null is superseded by, and no longer reported separately from, the full twelve-combination test below, which avoids fitting a formula to two known answers by calibrating each sensor’s threshold in units of that sensor’s *own* noise standard deviation – estimated leave-one-fold-out so a fold’s own noise draw never informs its own cutoff – and applying the same fixed critical value  $z$  (standard normal significance levels, not fit to this paper’s results) across every combination. The raw-MSE threshold is architecture- and sensor-adaptive by construction; the rule itself has zero free parameters tuned to this data.

Eleven of twelve combinations retain a significant favorable effect under the adaptive threshold, shrunk from the naive rule’s raw magnitude but not eliminated – conservatism mostly discards genuine, common signal rather than filtering noise, describing most tested combinations, not a special case. **GWNET/volume is the one combination where raising the threshold moves the result, from directionally-favorable-but-null (−0.46%,  $p = 0.356$ ) to nominally significant (−0.63%,  $p = 0.028$ ) – the reversal that motivated this investigation.** This table runs  $12 \times 4 = 48$

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TABLE 17

Adaptive noise-calibrated threshold vs. the naive rule ( $z=0$ ), all twelve (architecture, task) combinations with seed-control checkpoints available, 30 folds, volume and speed.  $z=1.645$  is a conventional one-sided 95% critical value, not tuned to this data.

| Model      | Task   | Naive ( $z=0$ ) $\Delta$ ( $p$ ) | Adaptive ( $z=1.645$ ) $\Delta$ ( $p$ ) |
|------------|--------|----------------------------------|-----------------------------------------|
| GMAN       | volume | −14.14% (5.3e-10)                | −6.06% (2.9e-7)                         |
| STID       | volume | −12.46% (1.4e-13)                | −5.56% (5.5e-10)                        |
| STGCN      | volume | −7.84% (4.9e-6)                  | −2.73% (8.3e-6)                         |
| STAEformer | volume | −4.49% (9.4e-4)                  | −1.61% (2.1e-3)                         |
| DCRNN      | volume | −4.06% (1.3e-3)                  | −1.38% (1.8e-3)                         |
| GWNET      | volume | −0.46% (0.356)                   | −0.63% (0.028)                          |
| GMAN       | speed  | −5.54% (9.6e-19)                 | −3.39% (5.8e-18)                        |
| STID       | speed  | −4.86% (8.2e-18)                 | −3.67% (1.4e-15)                        |
| STGCN      | speed  | −3.73% (4.4e-14)                 | −1.90% (2.6e-11)                        |
| GWNET      | speed  | −1.87% (1.6e-5)                  | −1.02% (1.8e-4)                         |
| STAEformer | speed  | −1.23% (1.7e-6)                  | −0.57% (1.3e-7)                         |
| DCRNN      | speed  | −1.51% (6.2e-7)                  | −0.67% (1.3e-7)                         |

significance tests, so whether the reversal is trustworthy is a multiple-comparisons question: under Bonferroni correction ( $\alpha = 1.0 \times 10^{-3}$ ), GWNET/volume does not survive at any  $z$  (the other eleven combinations do); under the less conservative Benjamini–Hochberg FDR control ( $q = 0.05$ ), standard for exploratory multi-comparison testing, it survives at all four  $z$  levels ( $p = 0.018$ – $0.043$ ) – consistent across thresholds rather than a single lucky draw, but explicitly *not* a confirmatory result. We report the Bonferroni-negative result alongside the FDR-positive one rather than showing only the more favorable correction.

In short, the threshold refines but does not reverse the underlying picture: for the eleven combinations with unambiguous real per-sensor signal it trades effect size for statistical caution; GWNET/volume is the sole combination where the choice of threshold changes the qualitative verdict, and only under the less conservative of the two corrections. The methodological contribution is modest but generalizes past the original 2-combination pilot with zero free parameters tuned to this data.

That a cheap, external, per-sensor selector recovers – and, for the one architecture already favorable, amplifies – the benefit in thirteen of fourteen combinations indicates that the capacity-gap account developed throughout this section explains why an architecture *cannot learn this gating internally from its own training signal*, not that the underlying sensor-level heterogeneity in when injection helps is itself capacity-dependent. As a concrete illustration for STAEformer/speed (the combination with the most severe uniform-injection harm, and 396 sensors, the largest sensor set tested), we additionally verify that this per-sensor heterogeneity is real rather than an artifact of finite-sample noise: treating each sensor’s 30-fold win/loss record as a draw from a shared binomial win rate (the pooled 38.2%) produces far less dispersion than observed – a chi-square overdispersion test rejects this null at  $4.03\times$  the expected variance ( $p \approx 0$ ) – and after Benjamini–Hochberg false-discovery-rate correction across all 396 per-sensor binomial tests, 76 sensors remain significant at  $q < 0.05$ : 31 significantly *helped* by injection and 45 significantly *hurt*. This split is itself evidence for, not against, a selective-gating approach: roughly a fifth of sensors have a real, individually detectable, and oppositely signed response to the same fixed injection mechanism.

**Robustness check: a random-selection negative control.** A natural objection is that mixing two independently trained checkpoints per sensor might improve accuracy for purely generic ensembling reasons, regardless of whether the selector’s validation-based choices are informative. We rule this out directly: for each fold and combination, we redraw 200 random per-sensor selection masks – matched to the exact number of sensors the real validation-based selector assigned to injection that fold – apply each to that fold’s test predictions, and compare the resulting distribution of mean %MSE changes to the real selector’s actual result. This control now covers all fourteen combinations in the main paper’s Table 12. In all fourteen combinations tested, the real, validation-informed result falls outside this null distribution entirely (empirical  $p = 0.0000$  against 200 random draws per fold, i.e. no random mask matched or exceeded the real result in any combination). The comparison is starkest for Graph WaveNet/volume: a random per-sensor mix is, on average, +8.2% *worse* than plain (unsurprising, since uniform injection is catastrophically harmful there and a random mix still injects into most sensors indiscriminately), while the validation-informed selector reduces this to a statistically indistinguishable −0.46%. A purely generic ensembling effect would have made random and validation-informed mixing perform comparably; the large, consistent gap confirms the selector is using genuine, sensor-specific information from the validation split, not benefiting from mixing per se.

Two limits on this result: the validation-based selector chooses a different sensor subset each fold (a mean of 43–291 sensors out of 121–396 depending on combination), so this is evidence for fold-dependent heterogeneity, not a small fixed set of universally favorable sensors; and the experiment answers a narrower, more practical question than the architectural generalization study above – not whether a model’s internal mechanism learns to use OD well, but whether OD carries real value at some sensors, detectable and exploitable with a cheap external check – which is immediately practical since it requires no architectural change, no retraining, and only one extra validation-window forward pass per already-trained checkpoint.

## APPENDIX C

### CROSS-DATASET REPLICATION: DOES THE CAPACITY LAW HOLD BEYOND SEOUL?

Every result so far comes from one city and, for the between-architecture test, one signal (routed OD) – a skeptical reading is that the law describes reimplementing fidelity on our own private dataset, not a real property of signal injection. We test this on three independent, standard, publicly benchmarked datasets – PEMS-BAY (325 sensors, Bay Area), METR-LA (207 sensors, Los Angeles, DCRNN release), PeMSD7 (228 sensors, STGCN release, weekdays only, 44 days) – using a harder test of the law: a pure calendar climatology carrying no forecast content, injected through the same ten architectures. All three are natively 5-minute; we aggregate to hourly and match Seoul’s  $P=Q=12$ , since an earlier native-resolution check (effectively 1h-history/1h-horizon) found the injection effect’s *sign itself* flips on PEMS-BAY between horizons (+8.5% harmful at 1h vs.  $-0.2\%$  at 12h), making horizon-matching necessary rather than optional. METR-LA is 8.1% missing at native resolution (vs. PEMS-BAY’s 0.003%); missing hours after aggregation (6.2%) are linearly interpolated per sensor rather than left as literal zeros, which we verified changes the result materially – the uninterpolated version shows a reversed, incoherent pattern the interpolated version does not.

TABLE 18

Capacity-law replication across four independent datasets: Spearman correlation between  $\log_{10}(\text{parameters})$  and injection effect across the same ten architectures, hourly resolution, horizon matched to Seoul (12h  $\rightarrow$  12h).

| Dataset            | Signal               | $\rho$        | $p$           |
|--------------------|----------------------|---------------|---------------|
| Seoul (this paper) | routed OD forecast   | <b>+0.891</b> | <b>5.0e-4</b> |
| PEMS-BAY           | calendar climatology | <b>+0.830</b> | <b>0.003</b>  |
| PeMSD7             | calendar climatology | +0.503        | 0.138         |
| METR-LA            | calendar climatology | +0.370        | 0.293         |

All four correlations are positive; under a null of no true effect, four independent draws landing on the same side is itself informative ( $p = 1/16$  by sign alone, before considering magnitude). Two reach significance at  $\alpha = 0.05$  (Seoul, PEMS-BAY); two do not (PeMSD7, METR-LA). We do not read this as the law failing to replicate in half the cases so much as replicating in direction everywhere and in significance only where data quality supports it: PeMSD7 has 44 days total (versus Seoul’s full multi-year range), and METR-LA remains the noisiest of the four datasets even after interpolation. We report all four exactly as measured rather than the two favorable ones, and regard the capacity-gap direction as reproducible across cities and signal types, with its statistical strength – not its existence – depending on the host dataset’s size and data quality.

## APPENDIX D

### QUALITATIVE CASE STUDY

To make the quintile-level diagnostic concrete, Figure 9 examines our own (lightweight-model) fold-0 predictions at two finer grains. Panel (a) plots *absolute error*, Traffic-Only vs. Traffic-Hybrid, for a single representative sensor (C-13,

one of the sensors with the largest mean improvement) across the *entire* test week, shaded green where Hybrid’s error is lower and red where Only’s is – deliberately not a cropped window, since an earlier best-2-of-7-days crop made the pattern look cleaner and more one-sided than it is. At the level of one sensor and one hour, the comparison is genuinely noisy: Hybrid has the lower error in 90 of the 168 hours shown (54%), not uniformly. What the panel does show clearly is that the week’s single largest error, by a wide margin, is a morning-commute spike on day 1 (07:00) where Hybrid’s error is roughly half of Only’s – consistent with the OD signal supplying an earlier warning of an imminent demand surge that pure traffic autocorrelation only detects after it is already underway – and that green shading visibly dominates at the week’s other large spikes, even though it does not dominate hour-by-hour everywhere. We do not read this one sensor’s noisy week as the evidence for the paper’s claims; it illustrates a mechanism whose real statistical support is the 30-fold aggregate (the main paper’s Table 4) and the 121-sensor, 23-fold aggregate diagnostic (the main paper’s Figure 8). Panel (b) aggregates the same error-reduction quantity across all 121 volume sensors, which is where the pattern becomes clean: mean absolute-error reduction from injection peaks sharply at 07:00–08:00 (morning commute, +42.9 and +40.2 vehicles/h respectively) and shows a smaller secondary peak at 17:00 (evening commute, +29.4 vehicles/h), while overnight hours (01:00–05:00) and mid-afternoon (13:00, 19:00, 21:00) show the smallest reductions ( $< 10$  vehicles/h). This hour-of-day pattern is a complementary, independent view of the same underlying phenomenon as the anomaly-quintile diagnostic: commute peaks are simultaneously the hours where OD movement most directly and predictably causes the resulting road traffic (a train of individually scheduled, purpose-driven trips converging on the same time window), and, in aggregate across many similar weekdays, a comparatively *typical*, repeating pattern rather than an anomalous one – which is exactly the regime the main paper’s Figure 8 identifies as where the injected signal is most reliable and most useful.

Together, the results above support a **capacity-gap** account of the generalization failure: road-routed OD injection supplies real information, but only information that a sufficiently expressive spatiotemporal architecture already extracts implicitly from its own inputs. The technique is valuable specifically where model capacity, not information availability, is the binding constraint on accuracy.

## APPENDIX E

### ADDITIONAL EXPLORATORY FINDINGS

Beyond the primary injection study, we report two further exploratory investigations that were not central to the paper’s core question but that may be useful to readers pursuing related directions.

#### E.1 Spatial Embedding Initialization (node2vec)

Learned adaptive-graph architectures such as AGCRN and GTS (the main paper’s Section 2.2) initialize their node embeddings randomly and let the forecasting objective shape them end-to-end. We tested whether initializing these

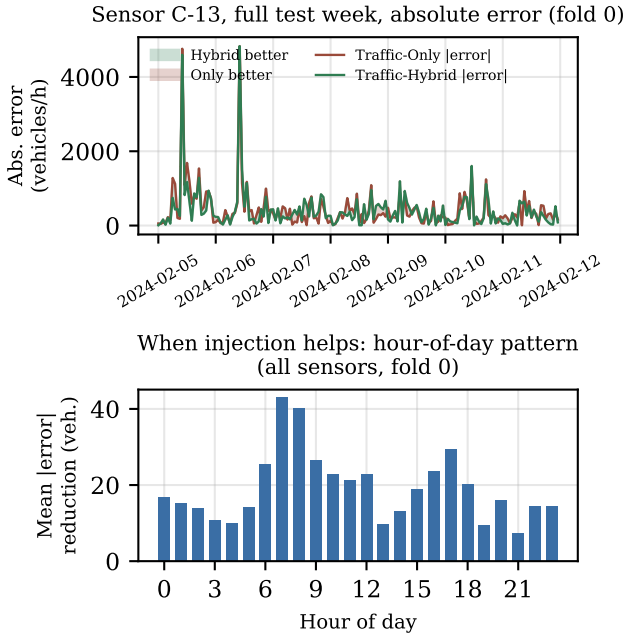


Fig. 9. Qualitative case study, fold 0: (a) one sensor's absolute error, full test week (not cropped) – green/red shading shows which variant wins each hour; noisy hour-to-hour (Hybrid wins 90/168 hours) but Hybrid clearly wins the week's largest spikes; (b) the same error-reduction quantity aggregated over all 121 sensors, where the commute-peak pattern is unambiguous.

embeddings from a node2vec [1] random-walk decomposition of the sensor graph – a purely structural prior, computed without reference to any traffic label – helps. We consider three regimes: random initialization (the standard baseline), a *frozen* node2vec initialization (the embedding is fixed at its node2vec value and never updated by gradient descent), and a *fine-tuned* node2vec initialization (the embedding starts at its node2vec value but is updated normally thereafter). Across the GMAN architecture, where this comparison was carried out most thoroughly, the frozen variant consistently underperforms random initialization, while the fine-tuned variant consistently outperforms it; we generalized the same fine-tuned-initialization mechanism to DCRNN, Graph WaveNet, and GTS via a shared `_init_node_emb` helper, though we did not pursue a full rolling-origin sweep of this direction to the same depth as the paper's central OD-injection study, and report it here as a smaller, secondary finding rather than a fully validated result. We view this result as broadly consistent with, and a useful independent data point for, the paper's central capacity-gap theme: a structural prior computed independently of the downstream task (node2vec's random-walk statistics, or, in the main study, the routed OD signal) is valuable as a *starting point or auxiliary signal* the model can adapt or selectively trust, but harmful when imposed rigidly (a frozen embedding, or, in Appendix A's ablation, an over-engineered fusion mechanism) in place of what the model is able to learn on its own given the freedom to do so.

## E.2 Purpose-Split OD Decomposition

Korean mobile-mobility records carry a `move_purpose` field distinguishing, among other categories, commute trips from other trip types. We built a purpose-split variant of the OD tensor that isolates the commute-purpose subset (after correcting a parsing bug in which the purpose codes were stored as float-formatted strings, e.g. "1.0", rather than the bare integers our initial filter expected, causing the commute subset to be silently empty until corrected; confirmed commute share after the fix is 15.5% of total OD volume) on the hypothesis that commute trips, being more temporally regular and route-predictable than discretionary trips, might be a more informative subset to route and inject than the full undifferentiated OD signal. We did not carry this direction through a full rolling-origin evaluation after the core research question was redirected toward the generalization study of the main paper's Section 8, and report the corrected, ready-to-use dataset and the motivating hypothesis here as a concrete direction for future work rather than as a completed result.

## REFERENCES

- [1] A. Grover and J. Leskovec, "node2vec: Scalable feature learning for networks," in *ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD)*, 2016.